CAMPAIGN RESULTS

Best Training Logging Validation Campaign Results Report

Best Training Logging Validation Campaign

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report summarizes the completed one-item campaign prepared in:

- `doc/reports/campaign_plans/2026-03-14-00-56-06_best_training_logging_validation_campaign_plan_report.md`

  The campaign objective was not to compare multiple model variants. The objective was to validate the revised live terminal behavior of `training/run_training_campaign.py` during a real feedforward training run using the current recommended preset:

- `config/training/feedforward/presets/best_training.yaml`

  The campaign completed successfully at the training and queue level:

- completed runs: `1`
- failed runs: `0`
- queue status for the campaign item: moved from `pending` to `completed`

  The execution artifact root is:

- `output/training_campaigns/2026-03-14-01-09-30_best_training_logging_validation_campaign_2026_03_14_00_56_06/`

Campaign Goal

The run was designed to answer four practical questions:

1. Does the campaign runner now show immediate startup output instead of the earlier silent initialization period?
2. Does the terminal now preserve the same readable sectioned training output used by `train_feedforward_network.py`?
3. Does the campaign still produce the expected queue, report, checkpoint, metric, and log artifacts?
4. Are there any residual logging issues still visible after the campaign-runner refactor?

Executed Configuration

Config	Source Preset	Run Name	Model Type	Campaign Role
`best_training_logging_validation`	`config/training/feedforward/presets/best_training.yaml`	`te_feedforward_best_training`	`feedforward`	Validate the updated live campaign logging on the current best practical preset

Training Outcome

The executed run reused the unchanged `best_training` hyperparameters and converged cleanly.

Config	Best Epoch	Wall Time	Val MAE [deg]	Val RMSE [deg]	Test MAE [deg]	Test RMSE [deg]
`best_training_logging_validation`	63	15.4 min	0.003171	0.003759	0.003579	0.004121

Interpretation

The training result is numerically stable and consistent with a valid feedforward run:

- the best checkpoint was selected at epoch `63`;
- early stopping stopped the run after epoch `76`;
- held-out validation remained strong with `val_mae = 0.003171 deg`;
- held-out test performance finished at `test_mae = 0.003579 deg` and `test_rmse = 0.004121 deg`.

  This confirms that the campaign runner changes did not break the core training, checkpointing, validation, or held-out test workflow.

Terminal Logging Validation Outcome

The live terminal behavior improved in the ways the campaign was supposed to validate.

Confirmed Improvements

- the campaign printed an immediate campaign header with `1/1` progress before training started;
- the full sectioned summary from the single-run training script remained visible:
  - configuration;
  - dataset summary;
  - model summary;
  - normalization statistics;
  - runtime summary;
  - training, validation, checkpoint evaluation, test, and artifact sections;
- epoch-level progress was readable directly in the active terminal during the real run;
- the campaign printed a clear completion footer with total completed and failed counts.

Campaign Artifact Validation

The queue-based workflow still produced the expected outputs:

Artifact	Status	Notes
`campaign_manifest.yaml`	generated	campaign-level machine-readable execution index
`campaign_execution_report.md`	generated	campaign-level execution summary
mirrored terminal log	generated	one log file stored under the campaign `logs/` folder
checkpoints	generated	best checkpoint and last checkpoint available
`training_test_metrics.yaml`	generated	final validation and test metrics available
`training_test_report.md`	generated	per-run markdown summary available

Residual Issues

The campaign also exposed two residual logging defects that should be treated as follow-up work rather than ignored.

1. Shutdown Exception After Successful Completion

After the campaign finished successfully, the terminal printed a final exception from `colorama` shutdown handling:

- `ValueError: I/O operation on closed file`

  The most likely cause is that the mirrored tee stream in the campaign runner is already closed when `colorama` executes its `atexit` reset callback. This does **not** invalidate the training results, but it means the campaign runner shutdown path is still not fully clean.

2. Mirrored Log File Is Not Yet Clean Enough

The stored mirrored campaign log still contains:

- ANSI escape sequences;
- mangled progress-bar glyphs in some progress sections.

  This means the live terminal experience improved substantially, but the serialized per-run log file is still not as clean as the direct terminal view.

Best-Performing Configuration

Because this campaign intentionally ran only one configuration, the best-performing configuration is trivially:

- `best_training_logging_validation`

  That does **not** mean this campaign re-selected the global best feedforward preset from scratch. It confirms that the already selected `best_training` preset still executes correctly through the revised campaign runner. From a workflow perspective, this is the main practical result of the campaign:
- the recommended preset remains usable inside the campaign queue flow,
- and the remaining issues are now concentrated in logging cleanup rather than in training correctness.

Main Conclusions

This one-item validation campaign supports four practical conclusions.

1. The New Live Campaign Logging Is Functionally Better

The original silence-at-start problem is no longer the dominant user experience. The campaign now shows immediate context and live training progress.

2. The Core Queue Workflow Remains Intact

The revised runner still completed the queue transition, training run, metrics export, checkpointing, and campaign reporting correctly.

3. The `best_training` Preset Remains Executable As A Campaign Item

The current recommended feedforward preset ran successfully inside the queue-based campaign workflow without breaking the training pipeline.

4. One Logging Fix Is Still Needed

The final `colorama` shutdown exception and the noisy mirrored log-file encoding show that the campaign-runner logging work is improved but not fully finished.

Recommended Next Improvements

The next useful follow-up tasks are

1. fix the runner shutdown path so `colorama` cannot write to a closed tee stream at process exit;
2. clean the mirrored log serialization so stored logs do not keep ANSI escape sequences or mangled progress glyphs;
3. re-run a short one-item validation campaign after that fix to confirm both live terminal output and stored logs are clean.

Post-Fix Rerun Addendum

After the original report was completed, the same one-item campaign was rerun after applying the shutdown fix documented in:

- `doc/technical/2026-03-14/2026-03-14-02-19-57_campaign_runner_colorama_shutdown_fix.md`

  The rerun completed successfully under:

- `output/training_campaigns/2026-03-14-02-22-52_best_training_logging_validation_campaign_2026_03_14_00_56_06/`

What Changed After The Fix

- the final `colorama` shutdown exception did **not** reappear in the rerun terminal output;
- the campaign still completed with `1` completed run and `0` failed runs;
- the live terminal behavior remained readable and consistent with the single-run training script;
- the stored mirrored log still needs cleanup because ANSI/progress serialization artifacts remain visible.

Updated Rerun Metrics

The shared run output directory `output/feedforward_network/te_feedforward_best_training/` now reflects the rerun artifacts rather than the original first-run artifacts.

Execution	Best Epoch	Wall Time	Val MAE [deg]	Val RMSE [deg]	Test MAE [deg]	Test RMSE
Original campaign run	63	15.4 min	0.003171	0.003759	0.003579	0.004121
Post-fix rerun	46	15.1 min	0.003039	0.003567	0.003409	0.003948

Interpretation Of The Addendum

This rerun closes one of the two residual issues from the original report:

- resolved: shutdown exception after successful completion;
- still open: mirrored log-file cleanliness.

It also shows slightly improved numerical results in the rerun, but those updated metrics should not be retroactively treated as the original campaign outcome. They belong to the post-fix confirmation run.

Rerun-Specific Artifact References

- `output/training_campaigns/2026-03-14-02-22-52_best_training_logging_validation_campaign_2026_03_14_00_56_06/campaign_manifest.yaml`
- `output/training_campaigns/2026-03-14-02-22-52_best_training_logging_validation_campaign_2026_03_14_00_56_06/campaign_execution_report.md`
- `output/training_campaigns/2026-03-14-02-22-52_best_training_logging_validation_campaign_2026_03_14_00_56_06/logs/2026-03-14-02-22-52_001_01_best_training_logging_validation.log`

Artifact References

Campaign-level references

- `output/training_campaigns/2026-03-14-01-09-30_best_training_logging_validation_campaign_2026_03_14_00_56_06/campaign_manifest.yaml`
- `output/training_campaigns/2026-03-14-01-09-30_best_training_logging_validation_campaign_2026_03_14_00_56_06/campaign_execution_report.md`
- `output/training_campaigns/2026-03-14-01-09-30_best_training_logging_validation_campaign_2026_03_14_00_56_06/logs/2026-03-14-01-09-30_001_01_best_training_logging_validation.log`

Run-level references:

- `output/feedforward_network/te_feedforward_best_training/best_checkpoint_path.txt` -> now points to the post-fix rerun checkpoint
- `output/feedforward_network/te_feedforward_best_training/training_test_metrics.yaml` -> now reflects the post-fix rerun metrics
- `output/feedforward_network/te_feedforward_best_training/training_test_report.md` -> now reflects the post-fix rerun summary
- `output/feedforward_network/te_feedforward_best_training/checkpoints/feedforward-epoch=063-val_mae=0.00317104.ckpt` -> original campaign best checkpoint

- `output/feedforward_network/te_feedforward_best_training/checkpoints/feedforward-epoch=046-val_mae=0.00303861.ckpt` -> post-fix rerun best checkpoint